PROJECT REPORT

ON

Impact of Artificial Intelligence on Education in Young Adults (15–25 Years)

submitted in partial fulfilment of the requirements for the award of degree of

BACHELOR OF ENGINEERING

In

COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

I hereby certify that the work which is being presented in the project report entitled **"**Impact of Artificial Intelligence on Education in Young Adults (15-25 Years)**"** in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of my own work carried out under the supervision of Dr. Gurpreet Singh. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura — Kanika (2210991735)

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This is to certify that the above statement made by the candidate is correct to the best of my knowledge and belief.

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Abstract

Artificial Intelligence has become an important technological advancement influencing educational practices among young adults aged 15–25 years. The increasing use of AI-based educational tools has transformed the way students learn, access information, and complete academic tasks. This research proposes a machine learning-based approach to analyze the impact of Artificial Intelligence on student academic performance using educational and behavioral data. The dataset includes factors such as AI usage hours, study hours, attendance percentage, previous academic scores, and AI dependency ratio. Data preprocessing techniques including data cleaning, feature selection, normalization, and dependency index calculation were applied before model implementation. Linear Regression was implemented to predict current student academic performance based on changing learning patterns influenced by AI technologies. The dataset was divided into 80% training data and 20% testing data. Model performance was analyzed through graphical visualization and prediction accuracy measures. Experimental results indicate that balanced AI usage combined with regular self-study and consistent attendance positively affects academic performance, while excessive dependency on AI may reduce learning consistency. The findings demonstrate that Artificial Intelligence can act as an effective educational support tool when used responsibly, although it should complement rather than replace independent learning and critical thinking abilities.

1. Introduction

Artificial Intelligence (AI) has emerged as one of the most influential technologies in recent years and has significantly transformed different sectors, especially education. AI-powered educational tools and intelligent systems have changed the traditional learning process by introducing personalized learning experiences, automated assessments, and smart academic support systems. With the rapid increase in digital learning technologies, students can now access educational content more efficiently and receive assistance beyond conventional classroom methods.

Young adults within the age group of 15-25 years are among the most active users of AI technologies for educational purposes. Students frequently use AI-based tools for answering questions, understanding difficult concepts, writing assignments, coding assistance, and academic preparation. Advanced platforms such as AI chatbots and intelligent tutoring systems provide quick responses and adaptive learning support, helping learners improve accessibility and convenience in education. The integration of AI into educational environments has changed study patterns and learning behaviors among students.

Despite its advantages, the increasing dependence on AI systems has raised concerns regarding learning balance and independent thinking abilities. Excessive reliance on AI tools may reduce self-study practices, critical thinking skills, and problem-solving abilities among students. Therefore, it is important to examine whether Artificial Intelligence functions as an effective educational assistant or contributes to dependency among learners. This research focuses on analyzing the impact of AI on education among young adults using machine learning techniques and evaluates how factors such as AI usage time, study habits, attendance, and previous academic performance influence student outcomes.

2. Objectives

- To study the impact of Artificial Intelligence on student performance
- To analyze the effect of AI dependency on learning behavior
- To predict academic performance using machine learning
- To evaluate the impact of study hours and attendance
- To promote balanced and responsible use of AI in education

3. Dataset Description

The dataset used in this project was created by analyzing "the impact of Artificial Intelligence on education among young adults aged 15-25 years". The dataset consists of 100 student profiles, where each record represents educational and learning behavior-related information used for predicting student academic performance.

Key Features Include:

- Age
- AI Usage Hours per Day
- Study Hours per Day
- Attendance Percentage
- Previous Academic Marks
- Current Academic Marks
- AI Dependency Index

4. Data Preprocessing

To ensure data quality and model performance, the following preprocessing steps were performed:

- Data cleaning and removal of unnecessary values
- Feature selection based on educational factors
- Calculation of AI Dependency Index
- Data normalization for better prediction accuracy
- Splitting dataset into:
    - 80% training data
    - 20% testing data

5. Methodology

This research uses a supervised machine learning approach for prediction and analysis of student academic performance.

Models Implemented:

• Linear Regression

The model uses educational and behavioural factors such as**:**

• AI Usage Hours

• Study Hours

• Attendance Percentage

• Previous Marks

• AI Dependency Index

It effectively identifies relationships between educational factors and predicts student academic performance based on learning behavior and AI usage patterns.

6. Model Evaluation Metrics

The models were evaluated using:

- **Actual vs Predicted Marks** — Comparison between predicted and actual student performance
- **Scatter Plot** — Relationships between AI usage and academic performance
- **Line Graph** — Shows trends and variations in student performance
- **Bar Graph** — Used to compare educational factors affecting results
- **Histogram** — Distribution of AI Dependency Index and learning behavior

7. Results and Findings

After implementing and evaluating the linear regression model, the following observations were obtained:

Key Observations:

- Balanced use of Artificial Intelligence along with regular self-study improved academic performance.
- Students with higher attendance and better previous academic scores showed more accurate prediction outcomes.
- Excessive dependency on AI reduced learning consistency and independent study behavior.
- The AI Dependency Index provided better insights into performance variations compared to AI usage alone.
- Artificial Intelligence was found to be more effective as a supportive educational tool rather than a replacement for self-study.

8. Feature Importance Analysis

The impact of educational features on student performance was analyzed using the Linear Regression model.

Key Influential Features:

- AI Usage Hours per Day
- Study Hours per Day
- Attendance Percentage
- Previous Academic Marks
- AI Dependency Index

9. Tools and Technologies

- **Programming Language:** Python
- **Libraries:** NumPy, Pandas, Scikit-learn, Matplotlib
- **Machine Learning Techniques:** Supervised Learning (Linear Regression)
- **Development Environment:** Jupyter Notebook / Google Colab
- **Model Training:** 80% Training Data and 20% Testing Data
- **Dataset Format:** CSV
- **Version Control (optional):** GitHub

10. Actual Code Snippet and Result

The following visual outputs were generated:

In [15]:
```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

In [16]:
```
# Load Dataset
df = pd.read_excel("AI_Education_100_Row_Dataset.xlsx")
```

In [17]:
```
# Display first 5 rows
print(df.head())
```

```
   Student_ID  Age  AI_Usage_Hours  Study_Hours  Attendance  Previous_Marks  \
0           1   15               1            3          75              65
1           2   25               3            4          96              63
2           3   20               5            7          71              80
3           4   23               3            6          83              83
4           5   22               2            7          78              85

   AI_Dependency_Index  Current_Marks
0                 0.33             64
1                 0.75             65
2                 0.71             82
3                 0.50             84
4                 0.29             90
```

Run Code

In [25]:
```
# Coefficients of factors
print("\nImpact of each factor:")
for feature, coef in zip(X.columns, model.coef_):
    print(feature, ":", coef)
```

```
Impact of each factor:
AI_Usage_Hours : 0.04646026426367864
Study_Hours : 0.005275760511108452
Attendance : -0.058716399783360185
Previous_Marks : 0.9757816672874013
AI_Dependency_Index : -0.5277165100058909
```

In [26]:
```
# Sample predictions
result = pd.DataFrame({
    'Actual Marks': y_test,
    'Predicted Marks': y_pred
})
print("\nPrediction Sample:")
print(result.head())
```

```
Prediction Sample:
    Actual Marks  Predicted Marks
83            65        68.524085
53            65        65.588604
70            79        75.918591
45            68        69.184570
44            91        86.829072
```

```
In [27]: # Chart 1: Actual vs Predicted
plt.scatter(y_test, y_pred)
plt.xlabel("Actual Marks")
plt.ylabel("Predicted Marks")
plt.title("Actual vs Predicted Academic Performance")
plt.show()
```

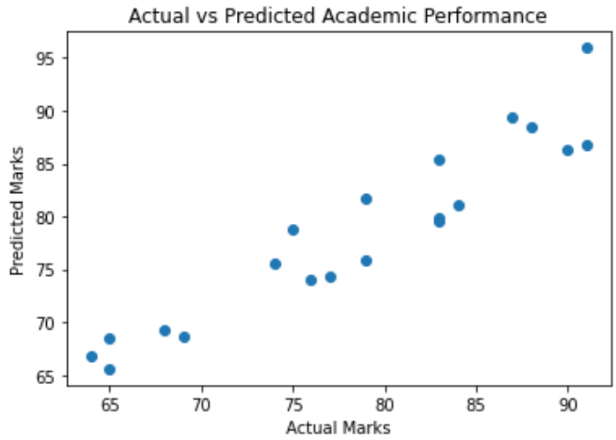

Figure 1: Actual vs Predicted Academic performance

```
In [28]: # Chart 2: AI Usage vs Current Marks
plt.scatter(df['AI_Usage_Hours'], df['Current_Marks'])
plt.xlabel("AI Usage Hours")
plt.ylabel("Current Marks")
plt.title("AI Usage Hours vs Academic Performance")
plt.show()
```

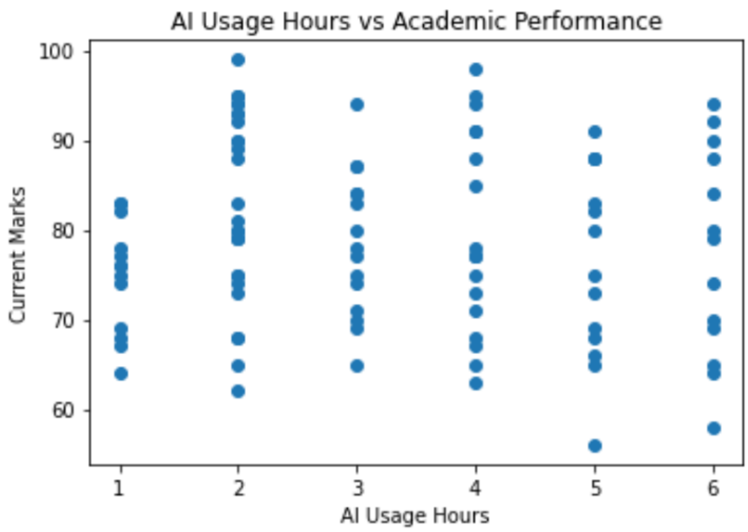

Figure 2: AI Usage vs Academic Performance

```
In [31]: # Attendance vs Academic Performance
plt.figure()
plt.scatter(df['Attendance'], df['Current_Marks'])
plt.xlabel("Attendance Percentage")
plt.ylabel("Current Marks")
plt.title("Attendance vs Academic Performance")
plt.show()
```

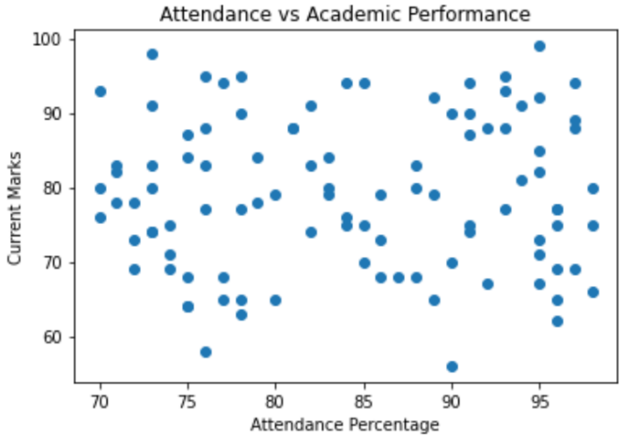

Figure 3: Attendance vs Academic Performance

```
In [30]: # AI Dependency Index Distribution
plt.figure()
plt.hist(df['AI_Dependency_Index'])
plt.xlabel("AI Dependency Index")
plt.ylabel("Number of Students")
plt.title("AI Dependency Index Distribution")
plt.show()
```

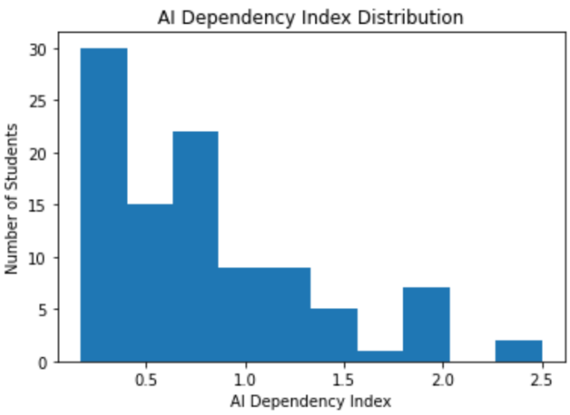

Figure 4: AI Dependency Index Distribution

11. Prediction Outcome

The final model predicts student academic performance based on educational and AI-related factors.

This prediction can be used as:

- Academic performance prediction
- Educational analytics
- Analysis of AI dependency on student learning

12. Conclusion

This research demonstrates that AI can improve learning efficiency and academic performance when used in a balanced manner along with regular self-study and consistent academic engagement. Factors such as attendance, study hours, and previous academic performance were found to play an important role in student outcomes.

The study also highlights that excessive dependence on AI technologies may negatively affect independent thinking and learning consistency. The proposed AI Dependency Index provided a better understanding of learning behavior and performance variation among students. Therefore, Artificial Intelligence should be used as a supportive educational tool rather than a replacement for self-study and critical thinking.

13. Future Scope

- Use real survey data from educational institutions.
- Increase the dataset size for better prediction accuracy
- Compare linear regression with neural network models
- Develop subject-wise prediction models
- Improve educational analytics using advanced AI techniques.

14. References

1. Holmes, W. Artificial Intelligence in Education
2. Luckin, R. Machine Learning and Human Intelligence
3. Kasneci, E. ChatGPT in Education
4. Ouyang, F. AI Research in Education

15. Appendices

Appendix A: Dataset Description

The dataset used in this research consists of 100 student profiles containing educational and behavioral factors related to learning patterns. The dataset includes variables such as age, AI usage hours, study hours, attendance percentage, previous marks, current marks and AI Dependency Index used for prediction and analysis.

Appendix B: Model Implementation

The model was implemented using Python and machine learning libraries including NumPy, Pandas, Matplotlib and Scikit-learn. Data preprocessing techniques such as data cleaning, feature selection and AI Dependency Index calculation were performed before model training.

Appendix C: Evaluation Metrics

The Performance of the model was evaluated by comparing actual and predicted student marks. Graphical analysis techniques such as scatter plots, line graphs, bar graphs and histograms were used to analyze relationships between AI usage, dependency and academic performance.